

4BUIS006C.2 Requirement Modelling

Group Coursework

Group A2

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1. Introduction

Contributors: Tharushi and Imdhadh

The University of Westminster is committed to fostering a sustainable future, and as part of this mission, the Green Westminster – Westminster Sustainability Engagement Platform has been proposed which aims to address key challenges in sustainability engagement, including raising awareness about available initiatives and maintaining long-term motivation for sustainable behaviors.

The aim of this project is to make a new platform for the University – **Green Westminster – Westminster Sustainability Engagement Platform** to encourage student and staff to adopt sustainable activities and increase their contribution with sustainability development.

Our team has been tasked with gathering, modeling, and defining the requirements for this platform, transforming the initial concept into a structured, actionable proposal.

To capture the requirements, we had to face an interview along with the tutors for the new platform and through the information obtained, we created a use case diagram to illustrate the functionalities of the application. A stakeholder onion model was developed to help identify and analyze the different groups of stakeholders involved in the project.

Furthermore, five different UML activity diagrams were drawn for five use cases identified in the use case diagram to provide a visual representation of the flow of the scenarios for each use case.

Scope of the Platform

The Platform will have the following features:

For User,

1. Sustainability education materials, News alerts and sustainability articles to keep users interested and informed, Leadership board
2. Carbon Tracking and Display overall Progress
3. Event listings and participation opportunities
4. Gamification elements like challenges and prizes
5. Tracking tools to gauge individual contributions

For Sustainability Coordinator,

1. Upload and Edit Sustainability education materials, News alerts and sustainability articles to keep users interested and informed, Leadership board
2. Edit and Upload Carbon Tracking impact Display overall Progress
3. Edit Event listings and participation opportunities
4. Add Gamification elements like challenges and prizes
5. Tracking tools to gauge individual contributions

Ultimately, we aim to help the University reduce its environmental impact and encourage a community of eco-conscious individuals.

2. Group Organization & Work Plan

Contributors: Imdhadh, Tharushi, Cleona, Samathi and Dahamsee

Group members,

Name	UOW no.	IIT no.
Imdhadh Iqbal	w2120848	20241316
Samathi Dikkumbura	w2120841	20230628
Dahamsee Kasthuriarachchi	w2120869	20231314
Cleona Fernando	w2120854	20241546
Tharushi Wijeratne	w2120855	20241418

Work Submission Timeline & Documentation

Date	Task Uploaded	Comments
29.01.2025	Initial project scope and interview questions	Shared with group for review
04.02.2025	Finalized interview scripts	Ready for execution
18.02.2025	User and system requirements	Submitted for feedback
25.02.2025	Use case diagrams	Draft for finalization
27.02.2025	Activity diagrams	Final version prepared
01.03.2025	Full project documentation	Ready for submission

Work plan,

Meeting Date, Time, Venue	Attendance	Details discussed / Decisions taken during the meeting	Each members' tasks before attending the meeting
29.01.2025 9.15am - 10.15am DB, Study area	Tharushi Imdhadh Cleona Samathi Dahamsee	- the scenario was explained to all the members and identified the scope for the new platform - Questions needed for the interview were discussed and selected a few sample questions.	Everyone went through the coursework specifications.

01.02.2025 11.05am - 12.15pm Online (Google Meet)	Tharushi Imdhadh Cleona Samathi Dahamsee	• The questions were filtered and discussed. • Distributed the roles among the members for the interview.	• Found appropriate questions for the interview and went through the specification.
04.02.2025 11 am - 12.30pm Online (Google Meet)	Tharushi Imdhadh Cleona Samathi Dahamsee	• Interview scripts were finalized along with the questions. • The question order and way of asking was discussed.	• Additional questions were brought, and they were discussed.
17.02.2025 8 pm - 9.50pm Online (Google Meet)	Tharushi Imdhadh Cleona Samathi Dahamsee	• The interview transcript was finalized. • The User requirements were compared and finalized.	• Each student bought user requirements
18.02.2025 11am - 12pm IIT DB study area	Tharushi Imdhadh Cleona Samathi Dahamsee	• The system requirements for the User requirements were taken. • All requirements were rechecked before it was presented for feedback.	
21.02.2025 1.30 pm - 3.20pm IIT Study area	Tharushi Imdhadh Cleona Samathi Dahamsee	• Use cases were finalized. • The user requirements were matched to the relevant use cases as well • Actors were taken and their relevant use cases were discussed.	• Use cases were taken by each student

25.02.2025 10am - 12pm IIT Study area	Tharushi Imdhadh Cleona Samathi Dahamsee	• The relationships among the actors and use cases were finalized. • Final sketch of the use case diagram was drawn.	• The relationships among the actors and use cases was bought by each student as a sample in order to help draft the final use case diagram.
27.02.2025 8.15 pm - 9.15 pm Online (Google Meet)	Tharushi Imdhadh Cleona Samathi Dahamsee	• All members went through the use case diagram and selected use cases to draw activity diagrams.	

3. Stakeholder Analysis

Contributors: Cleona, Imdhah and Samathi

3.1. Onion diagram

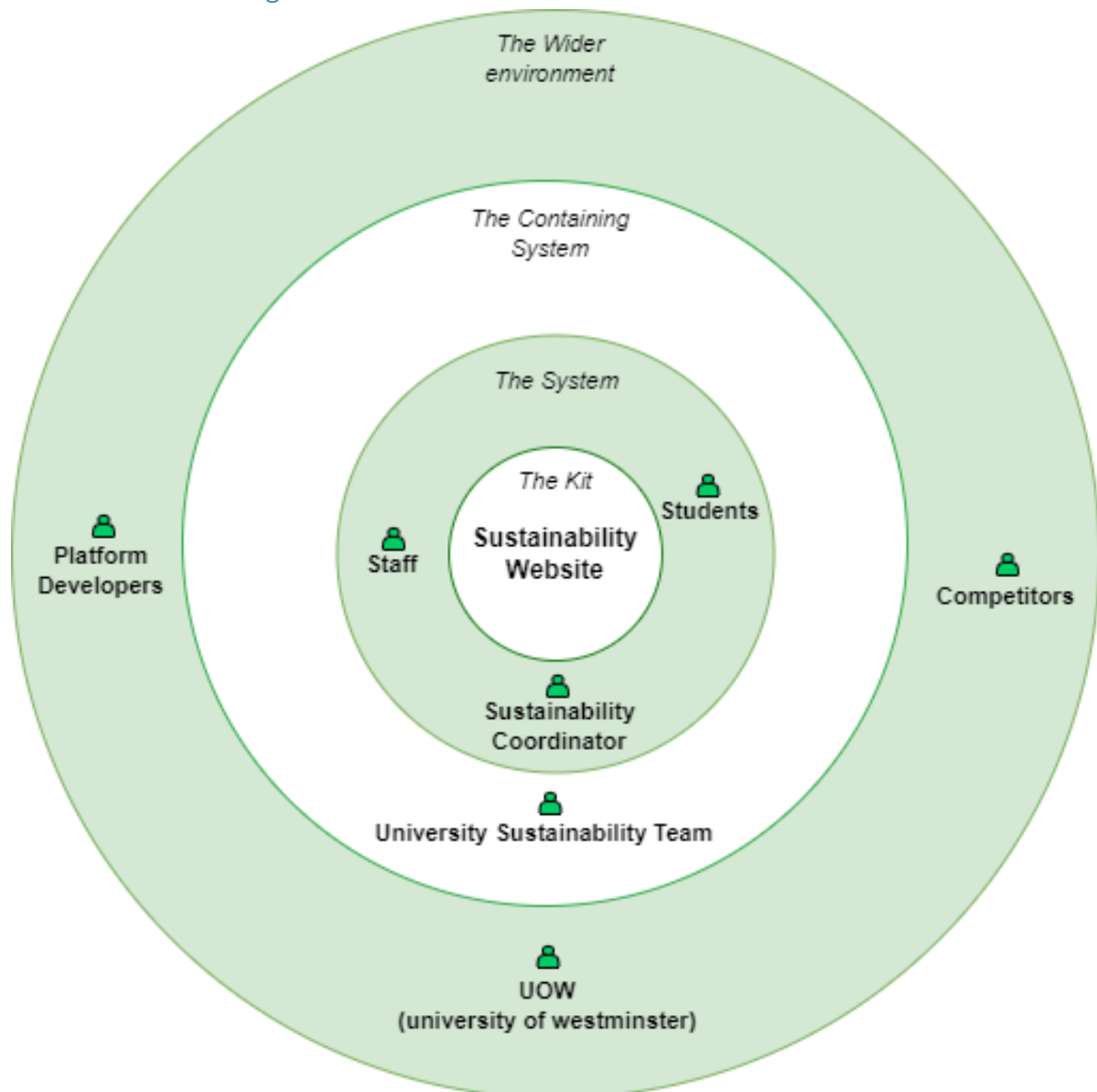


Figure 1 Onion Diagram

3.2. Stakeholder Descriptions

System layer

1. Staff

Interest - provide idea for the new platform that could help more students and staff get involved in eco-friendly actions and initiatives thereby, enhancing the University's overall impact on sustainability

2. Students

Interest - Primary users who will interact with the platform to learn, participate, and contribute to sustainability initiatives.

3. Sustainability Coordinator

Interest - Admins responsible for managing sustainability initiatives, monitoring platform engagement, and ensuring alignment with university policies, update events and challenges in the platform.

Containing system layer

4. University Sustainability Team

Interest - Oversees and supports the implementation of sustainability initiatives at the university level.

The wider environment

5. UOW (University of Westminster)

Interest - The broader institutional framework within which this platform operates.

6. Platform developers

Interest - Technical team responsible for designing, developing, and maintaining the platform.

7. Competitors

Interest – Other universities or other platforms having their own sustainability platforms which aims to attract students/staff or funding various initiatives.

4. Use case diagrams

Contributors: Imdhadh, Tharushi and Cleona

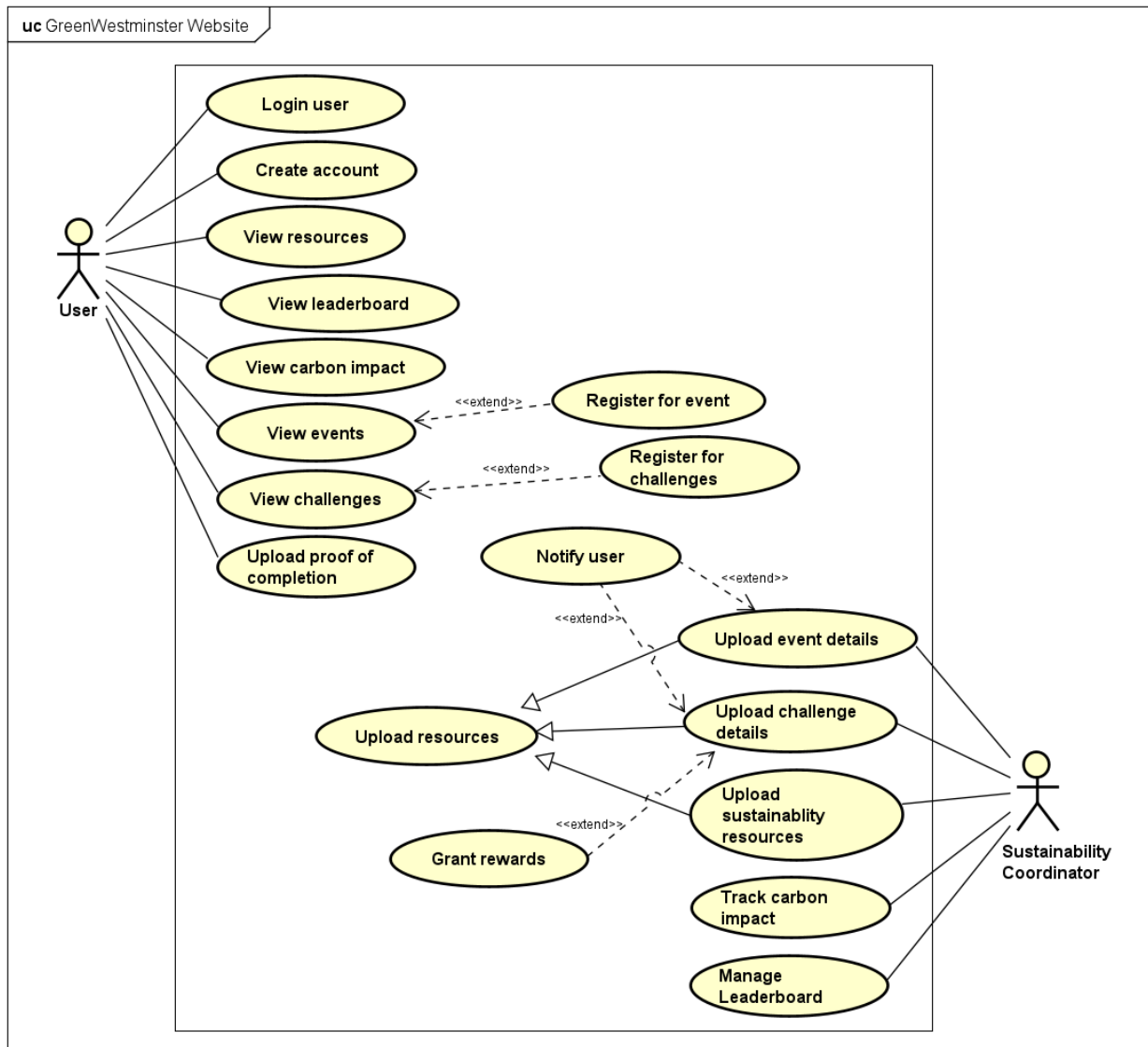


Figure 2 Use Case Diagram

5. List of Requirements

Contributors: Imdhadh, Tharushi and Cleona

5.1. Business Requirements.

1. **Promote Sustainability Awareness:** The platform will serve as a central hub to provide students and staff with easy access to sustainability-related information, initiatives, events, challenges and resources, ensuring they stay informed and engaged in eco-friendly actions.
2. **Encourage Active Participation:** By incorporating gamification elements such as challenges which award rewards, the platform will motivate users to adopt and sustain environmentally friendly behaviors both on and off campus.
3. **Support the University's Sustainability Goals:** The platform will contribute to the University of Westminster's commitment to sustainability by increasing student and staff engagement, helping drive measurable progress towards goals such as carbon neutrality.

5.2. User requirements and System requirements.

Req ID	Requirement	ReqType	Use case
UR1	UR1 - Users shall be able to log in using their University ID • SR1.1 The system shall authenticate users against the university's existing login credentials. • SR1.2 The system shall lock out accounts after three failed login attempts to prevent unauthorized access. • SR1.3 The system shall display an error message if authentication fails due to an incorrect ID or password.	FR FR NFR FR	Log into website
UR2	UR2 - Users shall be able to create a separate account in order to participate in events and challenges inside website.	FR	Create Account
UR3	UR3 – Users shall be able to view the events.	FR	View events
UR4	UR4 – Users shall be able to view the challenges	FR	View challenges
UR5	UR5 - Users shall be able to register for sustainability events	FR	Register For Events

	• SR3.1 The system shall display a list of upcoming sustainability events and challenges • SR3.2 The system shall allow users to click a "Register" button to sign up for an event • SR3.3 The system shall send a confirmation email upon successful registration.	FR FR FR	
UR6	UR6 - Users shall be able to register for sustainability challenges	FR	Register For Challenges
UR7	UR7 - Users shall be able to upload proof of participation in sustainability events & challenges. • SR4.1 The system shall provide an upload feature where users can submit images, PDFs, or documents as proof. • SR4.2 The system shall allow administrators to review and approve/reject submitted proofs. • SR4.3 The system shall notify users via email if their proof is approved or rejected.	FR FR FR FR	Upload Proof of completion
UR8	UR8 - Users shall view their progress, including carbon savings from actions. • SR5.1 The system shall display a dashboard showing users' accumulated carbon savings. • SR5.2 The system shall automatically update carbon savings after event participation is verified. • SR5.3 The system shall provide a visual progress bar to show a user's sustainability contributions.	FR FR FR FR	View carbon impact
UR9	UR9 - Users shall be able to view their rankings on leaderboard.	FR	View Leaderboard
UR10	UR10 – Users shall be able to view the sustainability-related resources.	FR	View resources
UR11	UR11 - The Sustainability Coordinator shall be able to upload event details • SR8.1 The system shall allow admins to upload educational materials, such as PDFs and videos. • SR8.2 The system shall provide an admin dashboard for creating and editing event details.	FR FR FR	Upload events

	SR8.3 The system shall send event notifications to users via email	FR	
UR12	UR12 - The Sustainability Coordinator shall be able to upload challenge details	FR	Upload Challenge
UR13	UR13 - The Sustainability Coordinator shall update sustainability-related resources.	FR	Update sustainability resources
UR14	UR14 - The Sustainability Coordinator shall track the carbon impact catalogue.	FR	Track carbon impact
UR15	UR15 – The Sustainability Coordinator shall manage leaderboard rankings.	FR	Manage Leaderboard
UR16	UR16 - The Sustainability Coordinator shall grant rewards for users after completion of challenges.	FR	Grant Rewards
UR17	UR17 - The Sustainability Coordinator shall notify users about events and challenges.	FR	Notify Users

6. UML Activity diagrams

6.1. Activity Diagram for View events.(Imdhadh Iqbal – W21208485 - 20241361)

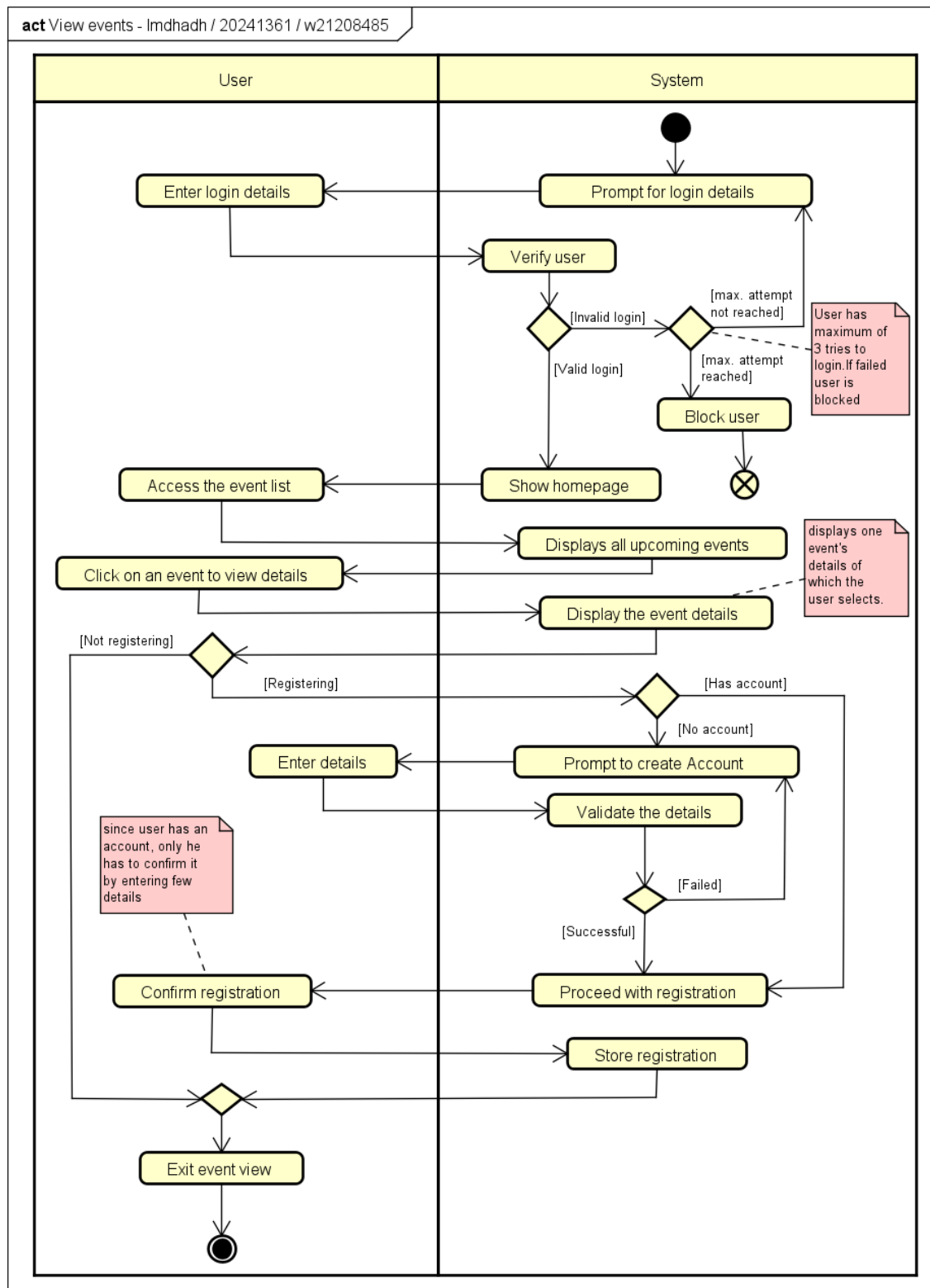


Figure 3 Activity Diagram for View events

6.2. Activity Diagram for View challenges.(Cleona - W2120854 - 20241546)

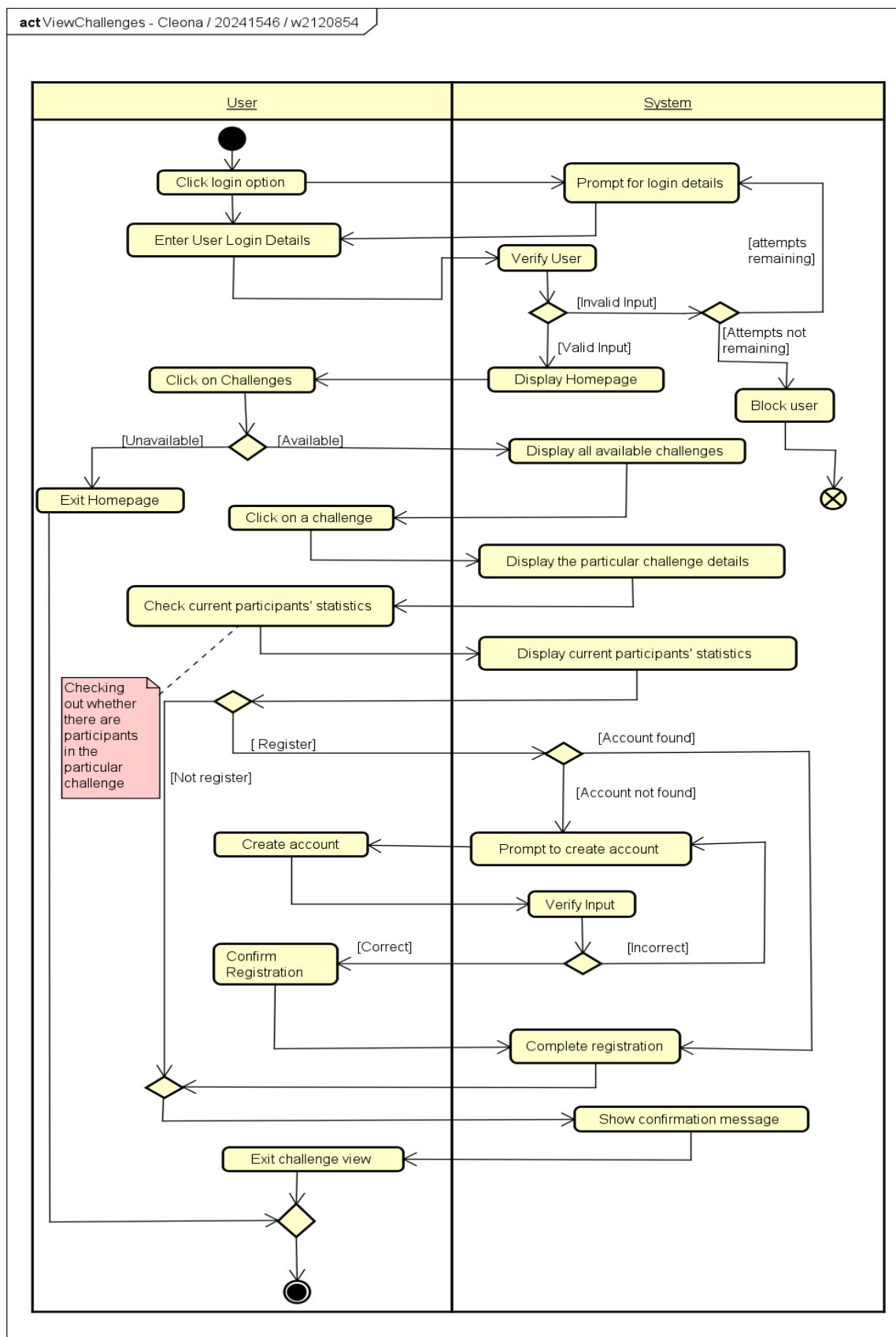


Figure 4 . Activity Diagram for View challenges

6.3. Activity Diagram for View carbon impact.(Dahamsee - W2120869 - 20231314)

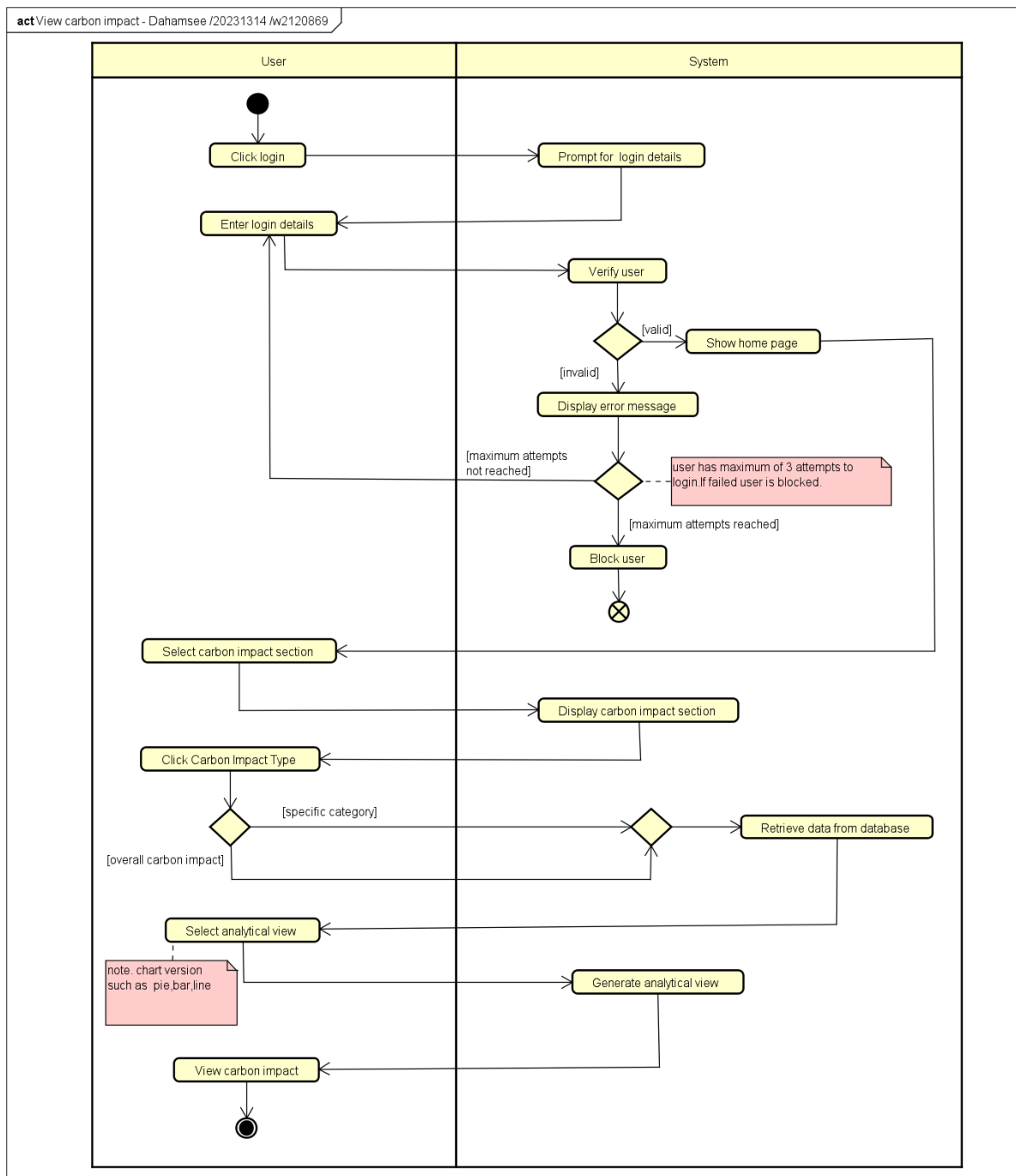


Figure 5 . Activity Diagram for View carbon impact

6.4. Activity Diagram for Upload sustainability resources.(Tharushi-W2120855 20241418)

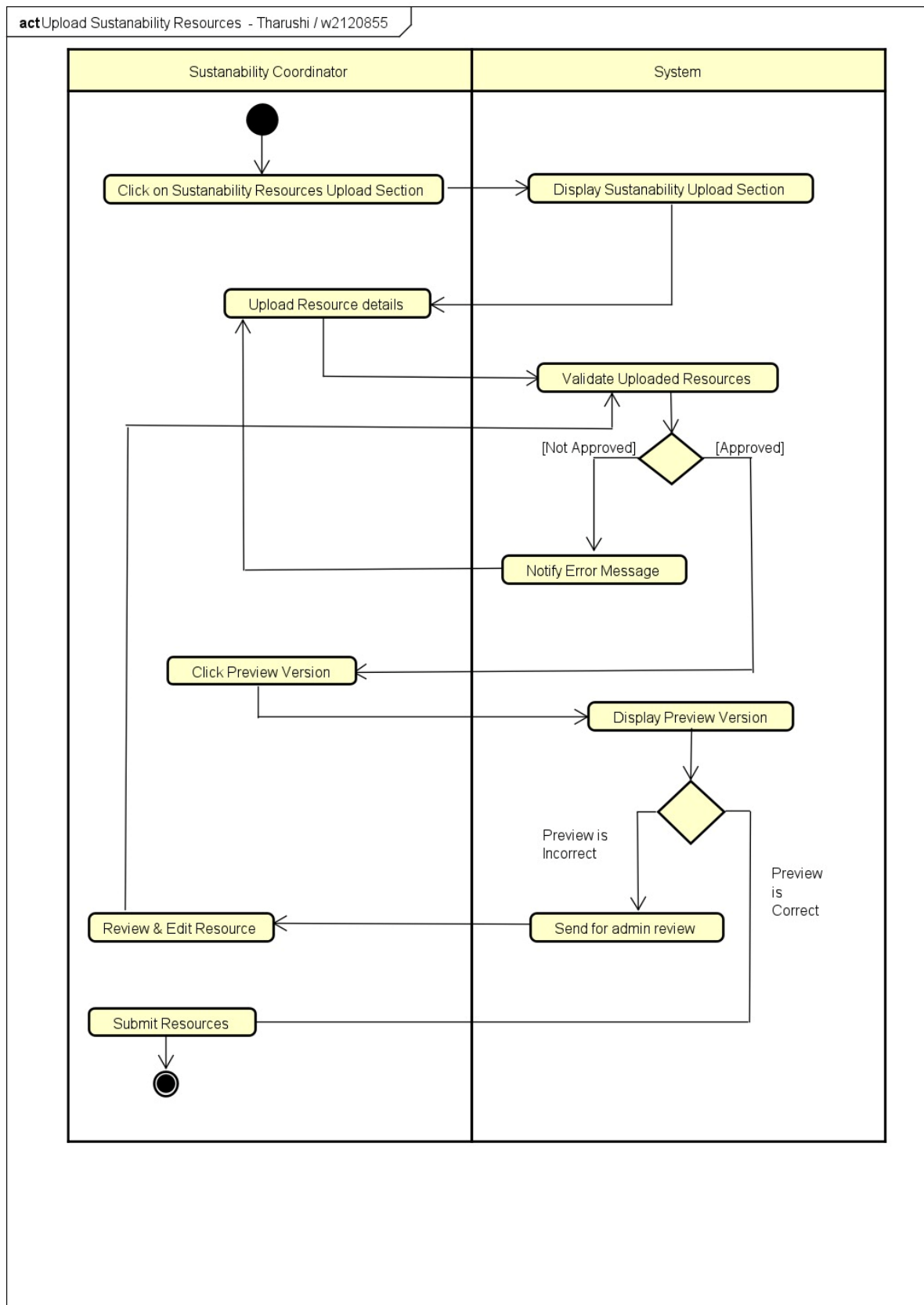


Figure 6 Activity Diagram for Upload sustainability resources

6.5. Activity Diagram for Upload event details.(Samathi – W21208416 - 20230628)

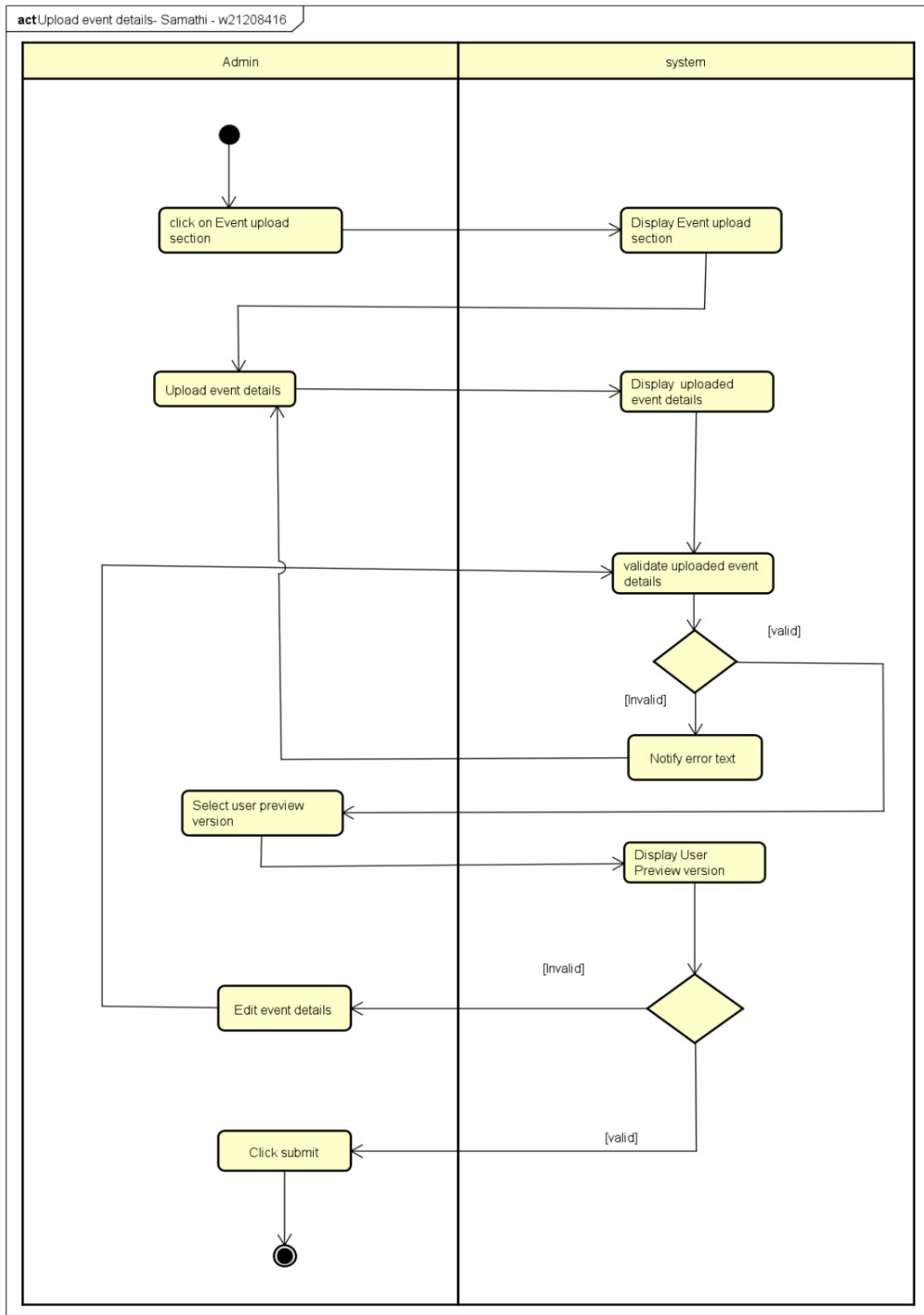


Figure 7 Activity Diagram for Upload event details

7. References

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APPENDIX

Appendix A – Draft / Earlier versions of Stakeholder Analysis

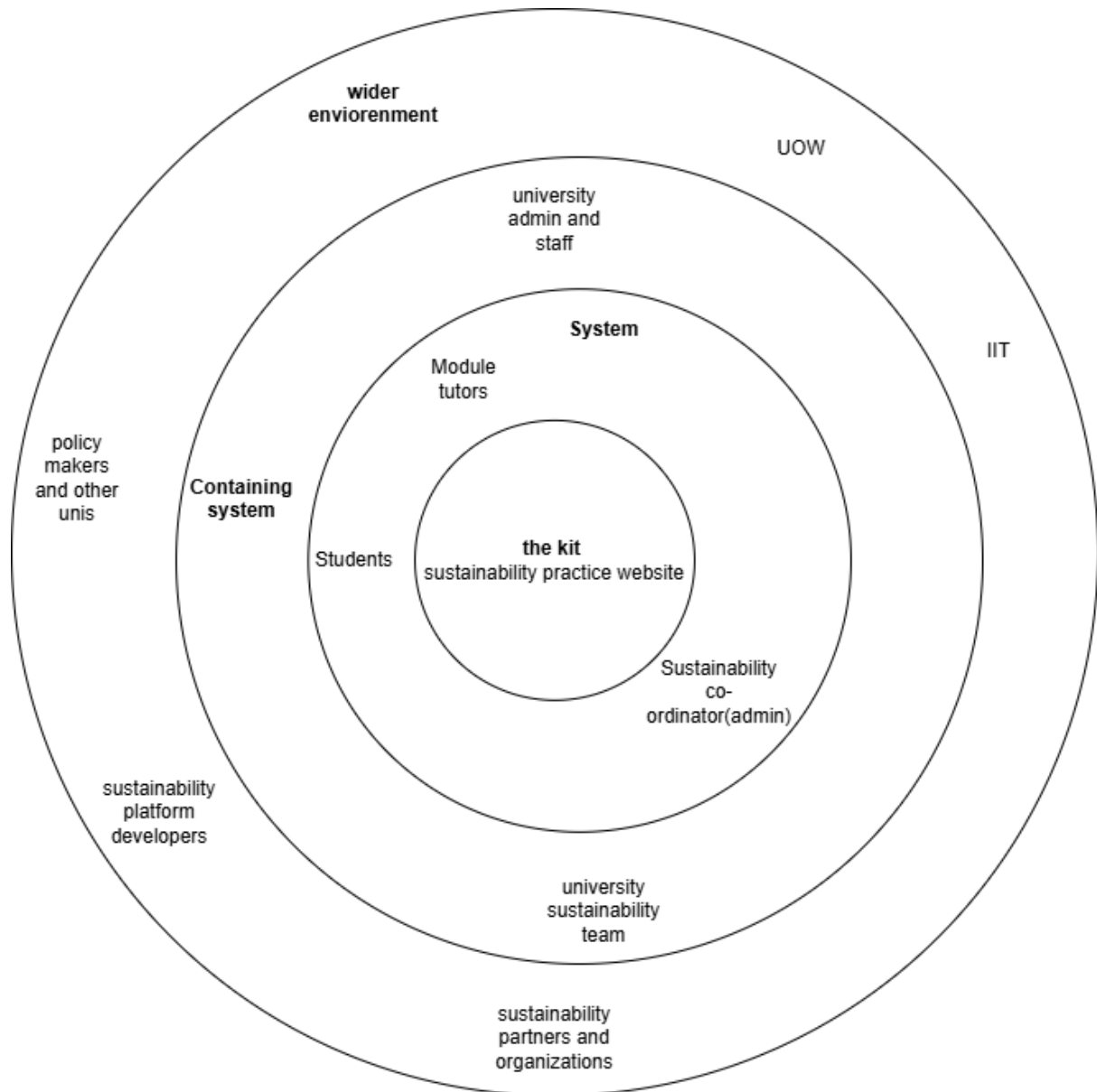


Figure 8 Earlier versions of Stakeholder Analysis

Appendix B – Draft / Earlier versions of Use case diagrams

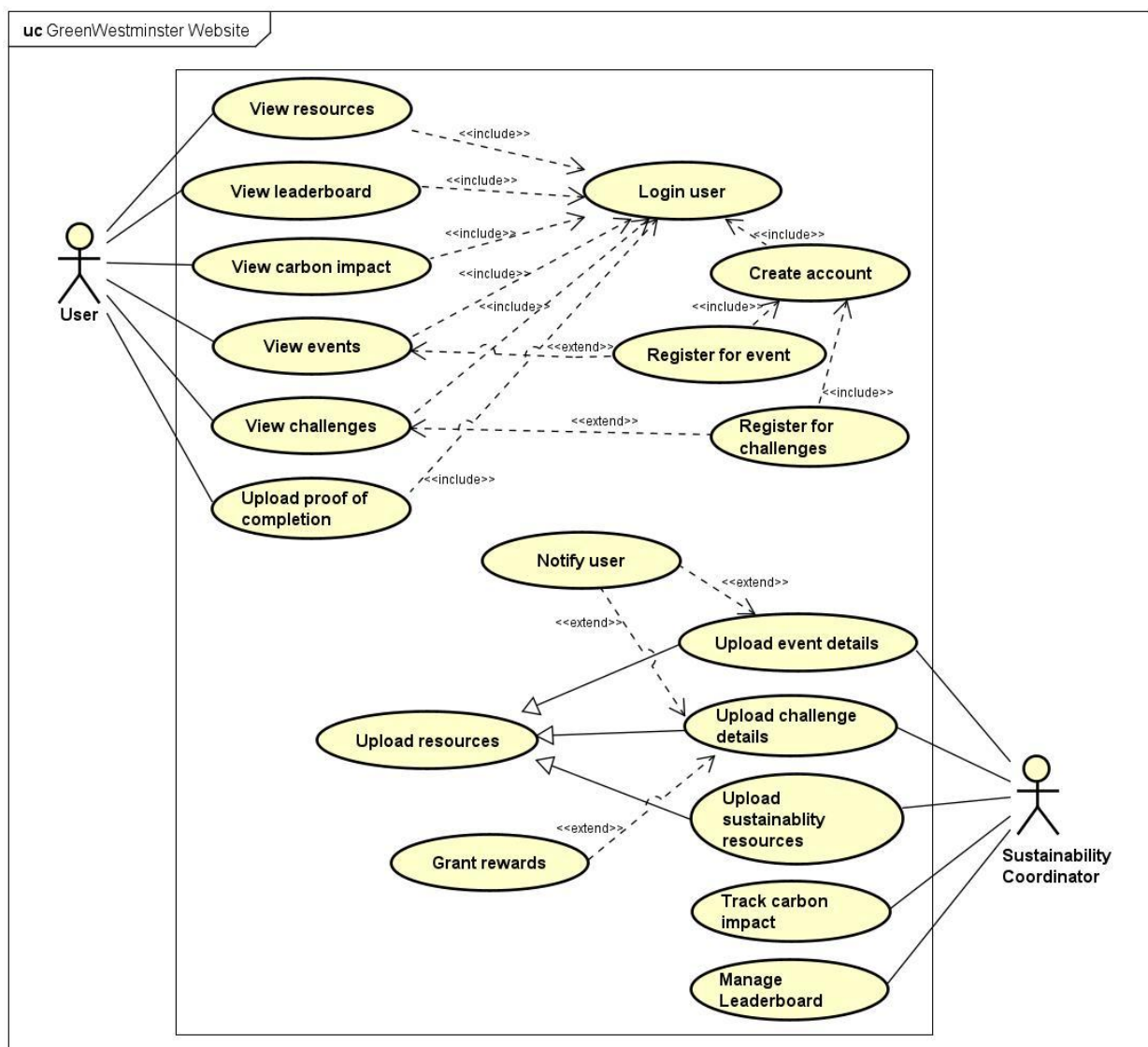


Figure 9 Earlier versions of Use case diagrams

Feedback and changes made.

"Create Account" and "Log In" were formerly featured as sub-use cases beneath a main use case in our use case diagram. But after getting input, we saw that categorizing them as sub-use cases complicated and made the graphic more difficult to read.

Instead of placing "Create Account" and "Log In" under another use case, we connected them directly to the user to make the diagram simpler. While still depicting the required user actions, this makes the diagram more understandable and straightforward.

Appendix C – Draft / Earlier versions of Requirements

Requirement ID	Requirement	Requirement Type	Use case
UR1	UR1 - Users shall be able to log in using their University ID • SR1.1 The system shall authenticate users against the university's existing login credentials. • SR1.2 The system shall lock out m after three failed login attempts to prevent unauthorized access. • SR1.3 The system shall display an error message if authentication fails due to an incorrect ID or password.	FR FR NFR FR	Log into website
UR2	UR2 - Users shall be able to create a separate account in order to participate in events and challenges in the website.	FR	Create Account
UR3	UR3 - Users shall be able to register for sustainability events • SR3.1 The system shall display a list of upcoming sustainability events and challenges • SR3.2 The system shall allow users to click a "Register" button to sign up for an event • SR3.3 The system shall send a confirmation email upon successful registration.	FR FR FR FR	Register For Events
UR4	Users shall be able to register for sustainability challenges	FR	Register For Challenges

UR5	UR5 Users shall be able to upload proof of participation in sustainability events SR4.1 The system shall provide an upload feature where users can submit images, PDFs, or documents as proof. SR4.2 The system shall allow administrators to review and approve/reject submitted proofs. SR4.3 The system shall notify users via email if their proof is approved or rejected.	FR FR FR FR	Upload Proof
UR6	UR6 - Users shall view their progress, including carbon savings from actions. • SR5.1 The system shall display a dashboard showing users' accumulated carbon savings. • SR5.2 The system shall automatically update carbon savings after event participation is verified. • SR5.3 The system shall provide a visual progress bar to show a user's sustainability contributions.	FR FR FR FR	View Progress
UR7	Users shall be able to view their rankings on leaderboard.	FR	View Leaderboard

UR8	UR10 - The IT administrator shall be able to upload event details • SR10.1 The system shall allow admins to upload educational materials, such as PDFs and videos. • SR10.2 The system shall provide an admin dashboard for creating and editing event details. • SR10.3 The system shall send event notifications to users via email	FR FR FR FR	Upload events
UR9	The IT administrator shall be able to upload challenge details	FR	Upload Challenge
UR10	The IT administrator shall update sustainability-related resources.	FR	Update sustainability resources
UR11	The IT administrator shall track the carbon impact catalogue.	FR	Track carbon impact
UR12	The IT administrator shall be able to verify proof of event participation.	FR	Verify Proof
UR13	The IT administrator shall manage leaderboard rankings	FR	Manage Leaderboard
UR14	The IT administrator shall grant rewards for users after completion of challenges.	FR	Grant Rewards
UR15	The IT administrator shall notify users about events and challenges	FR	Notify Users

Appendix D – Draft / Earlier versions of Activity Diagrams

Initial Activity Diagram for View events by Imdhadh

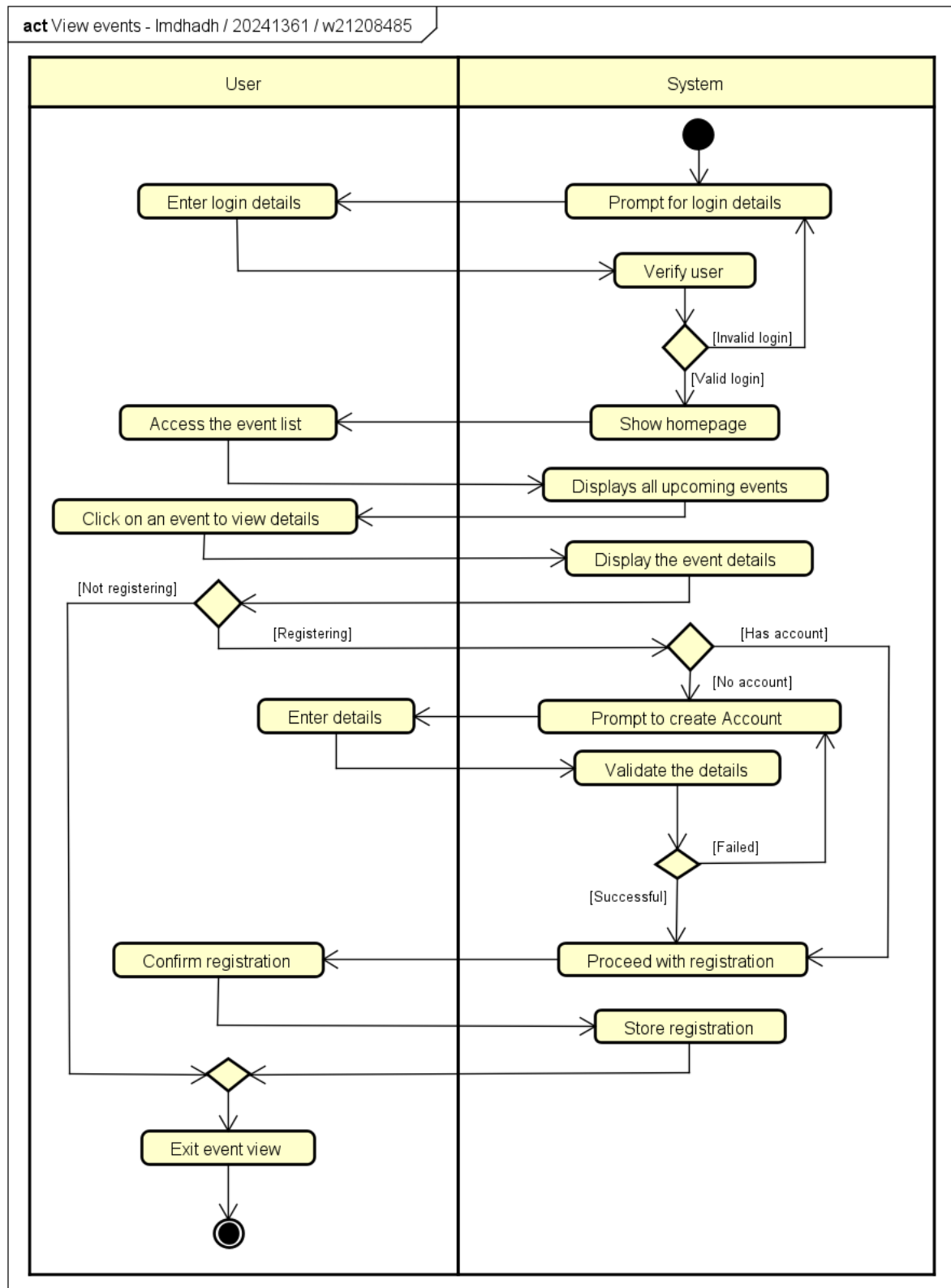


Figure 10 Earlier versions of Activity Diagram - View events

Feedback:

- Make maximum attempts for invalid login tries.
- Block user after 3 tries

Changes made:

- Added a decision box after invalid login and when user reached 3 tries, system blocks from retrying.

Initial Activity Diagram for View challenges by Cleona

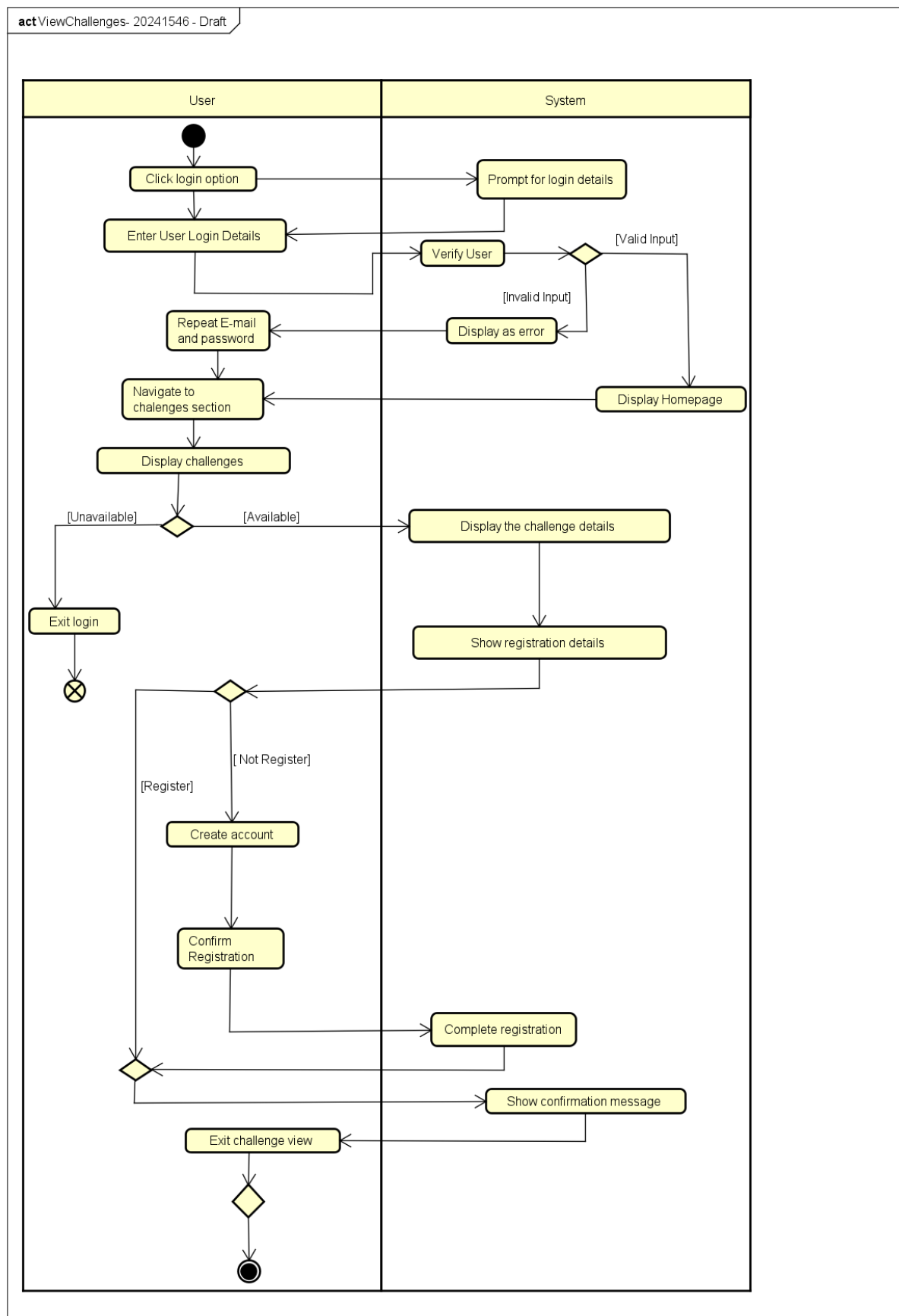


Figure 11 Earlier versions of Activity Diagram - View challenges.

Feedback:

- If there is an invalid output during login, add a decision there for more clarity.
- Make changes in Exit login.
- Make changes near display all challenges.
- The decision box was told to add near create account.

Changes made:

- Added a decision box (available attempts, unavailable attempts) and that was ended by a final flow node.
- Changed it to Exit Homepage and merged it to the merge node.
- The challenges section was elaborated, and more steps were added for easy understandability.
- When an account is not found, direct user to create account and validate the details (Here, the decision box was added).

Through feedback my final activity diagram was improved in error handling, user journey clarity, and registration logic.

Initial Activity Diagram for View carbon impact by Dahamsee

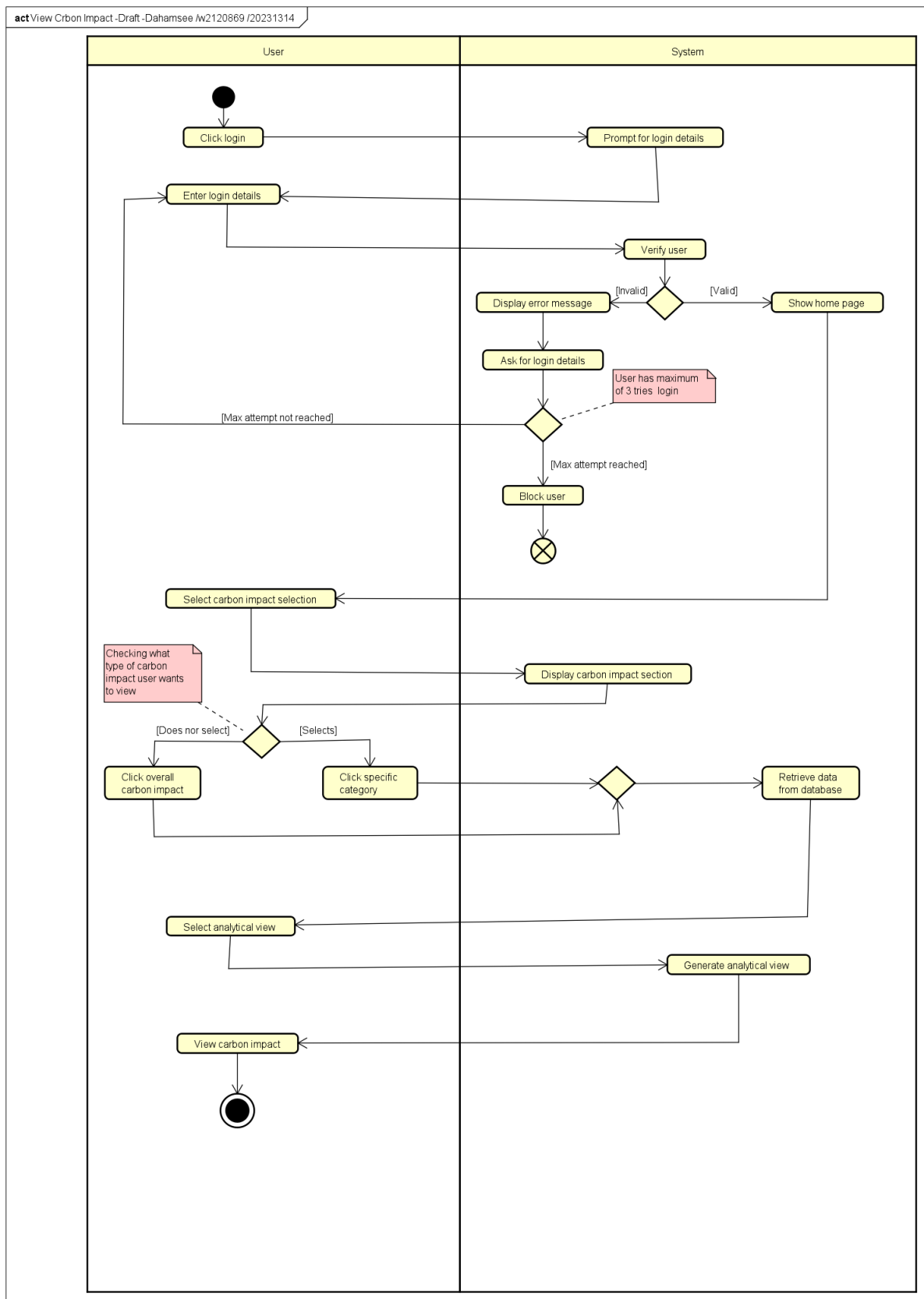


Figure 12 Earlier versions of Activity Diagram - View carbon impact

Feedback:

- Remove ask for login details
- Add an action called clicked carbon impact type
- Enter click overall carbon impact and click specific category into guard condition

Changes made:

- Removed ask for login details
- Added an action called click carbon impact type
- Entered click overall carbon impact and click specific category into guard condition

Initial Activity Diagram for Upload sustainability resources by Tharushi

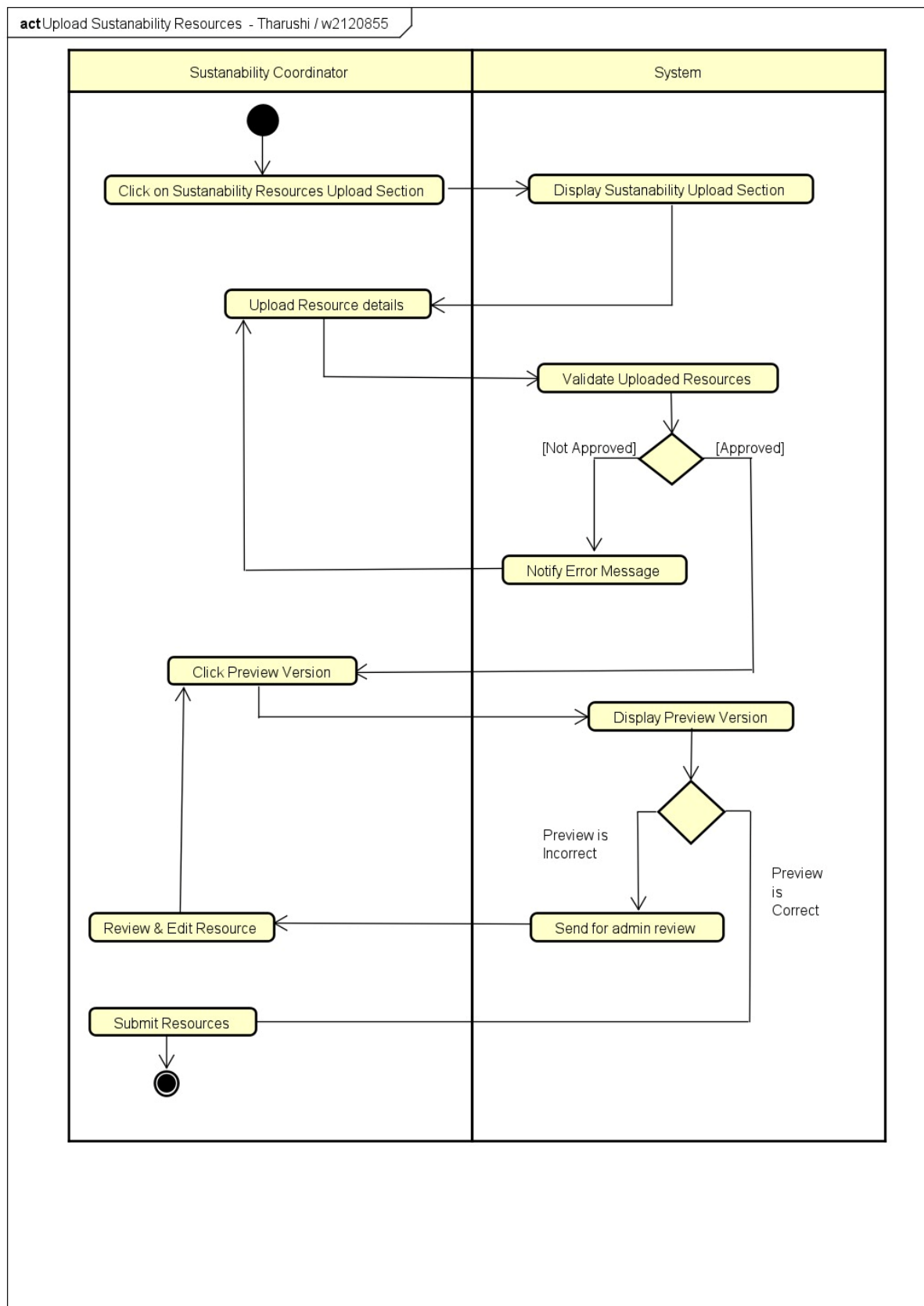


Figure 13 Earlier versions of Activity Diagram - Upload sustainability resources

Feedback:

- In this draft, the “Review & Edit Resource” action is connected to “Click preview version” action.
- This means the resource skips the validation step, so errors might not be checked properly. As a result, a resource that is not approved could move forward to preview, leading to mistakes in the final submission.

Changes made:

- Connect the "Review & Edit Resource" step back to "Validate Uploaded Resources" instead of "Click Preview Version."

Initial Activity Diagram for Upload event details by Samathi

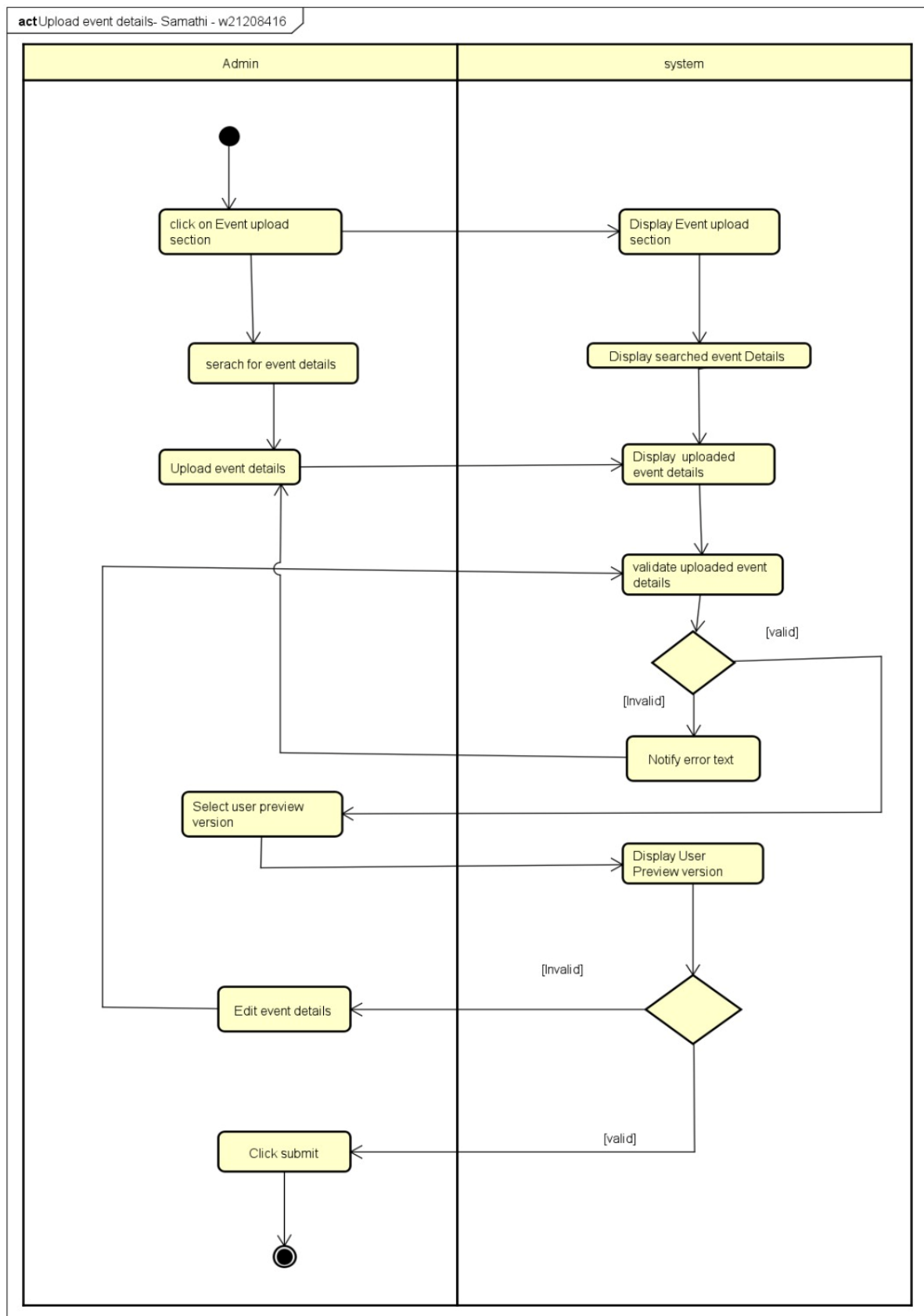


Figure 14 Earlier versions of Activity Diagram - Upload event details

Feedback:

- Remove search events
- Remove display searched event details
- Adjust the design

Changes made:

- Removed search events
- Removed display searched events
- Adjusted the design

Appendix E – Interview Transcript

Questions and Answers

1. Are there any existing systems that partially fulfill the sustainability needs?

Answer: Yes, there is an existing website that provides information about past and current sustainability projects, events, and challenges. However, many students and staff who want to participate don't know how to get involved. That's the main gap we want to address with this new platform.

2. Who else will be using the platform besides students and staff?

Answer: The Sustainability Coordinator will be responsible for updating the system with resources, articles, events, and the Carbon Impact Catalog. They ensure that activities' carbon impact is recorded and updated.

3. Should the platform be a mobile app or a website or should it be a part of an existing platform?

Answer: It should be a website with mobile compatibility.

4. Should the website be integrated into the university's existing system or be a standalone platform?

Answer: Standalone. Students and staff will log in with their university ID and password to view content. However, to participate in events or challenges, they must create a separate account on the platform. There will be no other integration with university systems.

5. Why do you need a new platform? What went wrong with previous sustainability efforts?

Answer:

The previous system didn't fail, but it didn't achieve the expected success. Many students didn't know how to sign up or where to go. There was no way to track participation or confirm if users completed events. The new platform will track progress, require proof of participation, and even offer rewards or recognition through leaderboards.

6. Will students and staff have the same role on the platform?

Answer: Yes, both students and staff will be able to participate in the same activities. However, admins will handle platform management.

7. What actions should users be able to perform on the platform?

Answer:

Register for events or challenges.

View progress and achievements.

See carbon savings from sustainability actions.

Upload proof of event completion.

Admins will update resources and track participation.

8. Will only carbon impact be tracked, or will other sustainability metrics be included?

Answer: Only carbon impact will be tracked. However, users must upload proof of event completion, which will be verified.

9. What are the accessibility requirements?

Answer: Students and staff can log in using their university ID to view content.

Leaderboards and other engagement features will only be accessible to those with valid university credentials. Once a student's university membership expires, they will lose access.

10. What are the special requirements for the platform (e.g., security, scalability, integration)?

Answer: The platform must follow the University IT Office's security guidelines.

11. Will external organizations or university clubs contribute to the platform?

Answer: University environmental clubs and associations may participate in events, but all content will be controlled by the admin.

12. Should there be a group chat or communication feature?

Answer: No, we are not planning to add that.

13. If you use the platform for six months, which feature do you expect to use the most?

Answer:

Carbon tracking, since it's a key focus.

Event participation tracking.

14. How can we keep users engaged? What do you think about increasing engagement?

Answer: Rewards will help. Some challenges may offer prizes, while others will recognize users through leaderboards.

15. Should there be feedback or surveys for students and staff?

Answer: No, we won't collect feedback through the platform. Communication will be via email notifications.

16. Are there any constraints for the platform?

Answer:

Budget: £20,000 for development.

Deadline: Must be ready by August 31st.